

HW5

1. (a) $P(G_{\text{father}}, G_{\text{mother}}, G_{\text{child}}) = P(G_{\text{father}}) P(G_{\text{mother}}) P(G_{\text{child}})$

则父、母、孩子基因相互独立，则只有网络(c)符合。

(b) $G_{\text{mother}}, G_{\text{father}} \rightarrow G_{\text{child}}, G \rightarrow H$

则(a)和用平遗传的假设一致。

(c)

G_{mother}	G_{father}	$P(G_{\text{child}}=l G_m, G_f)$	$P(G_{\text{child}}=r G_m, G_f)$
l	l	$1-m$	m
l	r	$\frac{1}{2}$	$\frac{1}{2}$
r	l	$\frac{1}{2}$	$\frac{1}{2}$
r	r	m	$1-m$

(d) $P(G_{\text{father}}=l) = P(G_{\text{mother}}=l) = q$

$P(G_{\text{father}}=r) = P(G_{\text{mother}}=r) = 1-q$

$$\begin{aligned}
 P(G_{\text{child}}=l) &= P(G_c=l | G_m=l, G_f=l) P(G_m=l) P(G_f=l) \\
 &\quad + P(G_c=l | G_m=l, G_f=r) P(G_m=l) P(G_f=r) \\
 &\quad + P(G_c=l | G_m=r, G_f=l) P(G_m=r) P(G_f=l) \\
 &\quad + P(G_c=l | G_m=r, G_f=r) P(G_m=r) P(G_f=r)
 \end{aligned}$$

$$= (1-m)q^2 + \frac{1}{2}q(1-q) + \frac{1}{2}(1-q)q + m(1-q)^2$$

$$= m + q - 2mq$$

2. ① 状态转移的条件概率分布表

E_t : 第 t 天是否睡眠充足, 若 $E_t=1$ 表示充足, $E_t=0$ 表示不充足

$$P(E_0)=0.7, P(\neg E_0)=0.3$$

E_{t-1}	$P(E_t=1 E_{t-1})$	$P(E_t=0 E_{t-1})$
1	0.8	0.2
0	0.3	0.7

② 观察序列 (R) 的条件概率分布表

R_t : 第 t 天是否红眼, 若 $R_t=1$ 表示红眼, $R_t=0$ 表示无红眼

E_t	$P(R_t=1 E_t)$	$P(R_t=0 E_t)$
1	0.2	0.8
0	0.7	0.3

③ 观察序列 (S) 的条件概率分布表

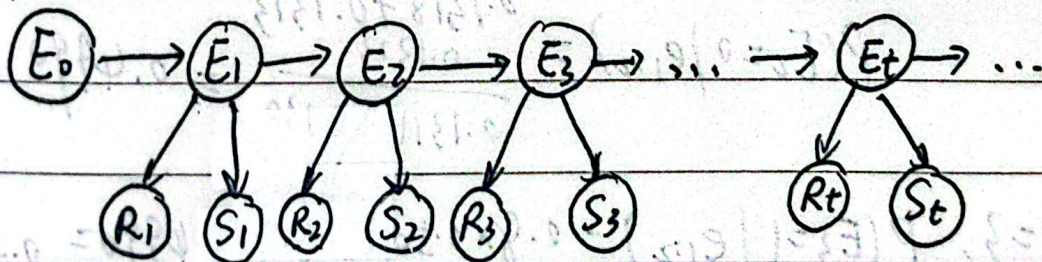
S_t : 第 t 天是否在课上睡觉, 若 $S_t=1$ 表示睡觉, $S_t=0$ 表示不睡觉

E_t	$P(S_t=1 E_t)$	$P(S_t=0 E_t)$
1	0.1	0.9
0	0.3	0.7

④ (R, S) 的条件概率分布表.

E_t	R_t	S_t	$P(R_t, S_t / E_t)$
1	1	1	$0.2 \times 0.1 = 0.02$
1	1	0	$0.2 \times 0.9 = 0.18$
1	0	1	$0.8 \times 0.1 = 0.08$
1	0	0	$0.8 \times 0.9 = 0.72$
0	1	1	$0.7 \times 0.3 = 0.21$
0	1	0	$0.7 \times 0.7 = 0.49$
0	0	1	$0.3 \times 0.3 = 0.09$
0	0	0	$0.3 \times 0.7 = 0.21$

贝叶斯网络:



3. (a)

$$\textcircled{1} t=1, P(E_1=1) = 0.8 \times 0.7 + 0.3 \times 0.3 = 0.65, P(E_1=0) = 0.35$$

$$\text{加入证据 } e_1: w(E_1=1) = 0.65 \times 0.72 = 0.468$$

$$w(E_1=0) = 0.35 \times 0.21 = 0.0735$$

$$\text{归一化: } P(E_1=1|e_1) = \frac{0.468}{0.468 + 0.0735} = 0.8643$$

$$P(E_1=0|e_1) = \frac{0.0735}{0.468 + 0.0735} = 0.1357$$

$$\textcircled{2} t=2, P(E_2=1|e_{1:2}) = 0.8 \times 0.8643 + 0.3 \times 0.1357 = 0.7321$$

$$P(E_2=0|e_{1:2}) = 0.2 \times 0.8643 + 0.7 \times 0.1357 = 0.2679$$

$$\text{加入证据 } e_2: w(E_2=1) = 0.7321 \times 0.18 = 0.1318$$

$$w(E_2=0) = 0.2679 \times 0.49 = 0.1313$$

$$\text{归一化: } P(E_2=1|e_{1:2}) = \frac{0.1318}{0.1318 + 0.1313} = 0.5010$$

$$P(E_2=0|e_{1:2}) = \frac{0.1313}{0.1318 + 0.1313} = 0.4990$$

$$\textcircled{3} t=3, P(E_3=1|e_{1:3}) = 0.8 \times 0.5010 + 0.3 \times 0.4990 = 0.5505$$

$$P(E_3=0|e_{1:3}) = 0.2 \times 0.5010 + 0.7 \times 0.4990 = 0.4495$$

$$\text{加入证据 } e_3: w(E_3=1) = 0.5505 \times 0.2 = 0.1101$$

$$w(E_3=0) = 0.4495 \times 0.21 = 0.0944$$

$$\text{归一化: } P(E_3=1|e_{1:3}) = \frac{0.1101}{0.1101 + 0.0944} = 0.1045$$

$$P(E_3=0|e_{1:3}) = \frac{0.0944}{0.1101 + 0.0944} = 0.8955$$

(b) $t=2$. 则求 $P(E_2 | e_{1:3})$

由(a)知 $P(E_2=1 | e_{1:2}) = 0.5010$, $P(E_2=0 | e_{1:2}) = 0.4990$

$E_2=1$ 时, $P(e_3 | E_2=1) = 0.02 \times 0.8 + 0.21 \times 0.2 = 0.058$

$E_2=0$ 时, $P(e_3 | E_2=0) = 0.02 \times 0.3 + 0.21 \times 0.7 = 0.153$

则 $W(E_2=1) = 0.5010 \times 0.058 = 0.0291$

$W(E_2=0) = 0.4990 \times 0.153 = 0.0763$

归一化: $P(E_2=1 | e_{1:3}) = \frac{0.0291}{0.0291 + 0.0763} = 0.2757$

$P(E_2=0 | e_{1:3}) = \frac{0.0763}{0.0291 + 0.0763} = 0.7243$